# FUNDAMENTALS OF MACHINE LEARNING

## (30 MINUTES)



Machine Learning (ML) is a subset of artificial intelligence that enables computers to learn from data and make predictions or decisions without being explicitly programmed for every scenario. Instead of following a predetermined set of rules, ML systems identify patterns, relationships, and structures in data and use these patterns to make intelligent decisions about new, unseen data. The ML life cycle follows a core workflow: **training** (learning patterns from data) and **inference** (eg. making predictions), supported by frameworks that streamline development.

### Learning objectives

After completing this module, you will be able to:

- Describe core fundamentals of machine learning
- Differentiate AI, ML and Deep Learning
- Useful of Machine Learning Framework

### 1.0 Introduction

Machine learning is an understanding on how machines learn requires a foundational knowledge of how data is processed, how models evaluate performance, and how learning approaches are categorized. Learning approaches consists:

- Supervised Learning: The model trains on labeled data (where both input and expected output are known). It is used for predicting values or categories.
- Unsupervised Learning: The model evaluates unlabeled data to find hidden structures, clusters, or anomalies.
- Reinforcement Learning: The model learns through trial and error, adjusting its actions based on a system of rewards and penalties.

Machine learning tasks are typically organized into three foundational categories based on the type of problem being solved and the nature of the available data. Each task serves a distinct purpose in extracting value from data.

### Classification

Classification is a *supervised learning* task that predicts a **discrete categorical label** (a class) for a given input. The model learns from labeled training data and then assigns new, unseen data points to one of several predefined categories. The algorithm maps input features to one of N possible categories by learning decision boundaries that separate the classes in feature space.

Common algorithms are Logistic Regression, Decision Trees and Random Forests, Support Vector Machines (SVM), K-Nearest Neighbors (KNN), Naive Bayes, and

Neural Networks. Real-world examples such as Email filtering (spam vs. inbox), Medical diagnosis (malignant vs. benign tumor), Credit card fraud detection, Sentiment analysis (positive, negative, neutral), and Image recognition (cat, dog, bird)

### Regression

Regression is a *supervised learning* task that predicts a **continuous numerical value** based on input features. Unlike classification, the output is a real number on a continuous scale, not a category. The algorithm learns a mathematical function that maps input variables to a continuous output, typically by minimizing the error between predicted and actual values.

Common algorithms are Linear Regression, Polynomial Regression, Ridge and Lasso Regression, Decision Tree Regression, Random Forest Regression, Gradient Boosting (XGBoost, LightGBM), and Neural Networks. Real-world examples such as Predicting house prices based on size, location, and amenities, Forecasting stock prices or sales revenue, Estimating temperature, rainfall, or weather conditions, Predicting customer lifetime value, and Estimating insurance risk and premiums

### Clustering

Clustering is an *unsupervised learning* task that **groups similar data points together** based on their features, without using pre-labeled training data. The algorithm discovers inherent structures or patterns in the data on its own. The algorithm measures similarity (often using distance metrics like Euclidean distance) and partitions data into clusters where points within a cluster are more similar to each other than to points in other clusters.

Common algorithms are K-Means Clustering, Hierarchical Clustering (Agglomerative / Divisive), DBSCAN (Density-Based Spatial Clustering), Mean Shift, Gaussian Mixture Models (GMM), and Spectral Clustering. Real-world examples such as Customer segmentation for targeted marketing, Document or news article grouping, Anomaly detection (outliers form their own cluster), Gene sequence analysis in bioinformatics, Image compression and color quantization, and Social network analysis (community detection).

Comparison of Primary Machine Learning Tasks

| Aspect | Classification | Regression | Clustering |
|---|---|---|---|
| Learning type | Supervised | Supervised | Unsupervised |
| Output type | Discrete categories | Continuous numbers | Group memberships |
| Requires labels | Yes | Yes | No |
| Goal | Predict a class | Predict a value | Discover structure |
| Key challenge | Class imbalance | Overfitting, outliers | Choosing K (number of clustering) |
| Example | Spam detection | House price prediction | Customer segmentation |

### 2.0 AI vs ML vs Deep Learning

Here's where many people get confused. These three terms are not synonymous, they are nested layers of the same technological stack. Let me break this down clearly.

**Artificial Intelligence (AI)**

Artificial Intelligence is any technique that enables machines to mimic human intelligence. It's the umbrella concept covering any technology that makes computers "smart." AI encompasses everything from simple rule-based systems to the most advanced neural networks. AI systems use various techniques such as rules-based (*if this, then that*), some use optimization algorithms, some use learning from data. The key is that the computer exhibits intelligent behavior. AI is like the entire field of "smart machinery." Anything that seems intelligent is AI.

Real world example:

- Siri: Understanding your voice commands and responding conversationally
- ChatGPT: Generating human-like text responses
- Self-driving cars: Navigating traffic and making driving decisions
- Game-playing AI: Winning at chess or Go against world champions
- Recommendation systems: Netflix suggesting your next show
- Spam filters: Deciding if an email is junk

**Machine Learning (ML)**

Machine Learning is a subset of AI that specifically focuses on systems that learn from data without being explicitly programmed. If a system learns from experience and improves its performance over time, it's machine learning. Narrower than AI but more practical for most real-world applications. ML powers most of the "intelligent" things you interact with daily.

Not all AI is machine learning. A rule-based expert system (like old medical diagnosis systems) is AI but not ML. However, all ML is AI. ML is AI that specifically learns from data. It's the "learning from experience" part of intelligence.

Real world example:

- Email spam filters: Learn which emails are likely junk based on patterns
- Product recommendations: Amazon learns your preferences from your browsing and purchase history
- Credit card fraud detection: Banks learn patterns of fraudulent transactions
- Predictive text: Your phone learns your typing patterns
- Disease diagnosis: Medical systems learn to identify diseases from medical imaging

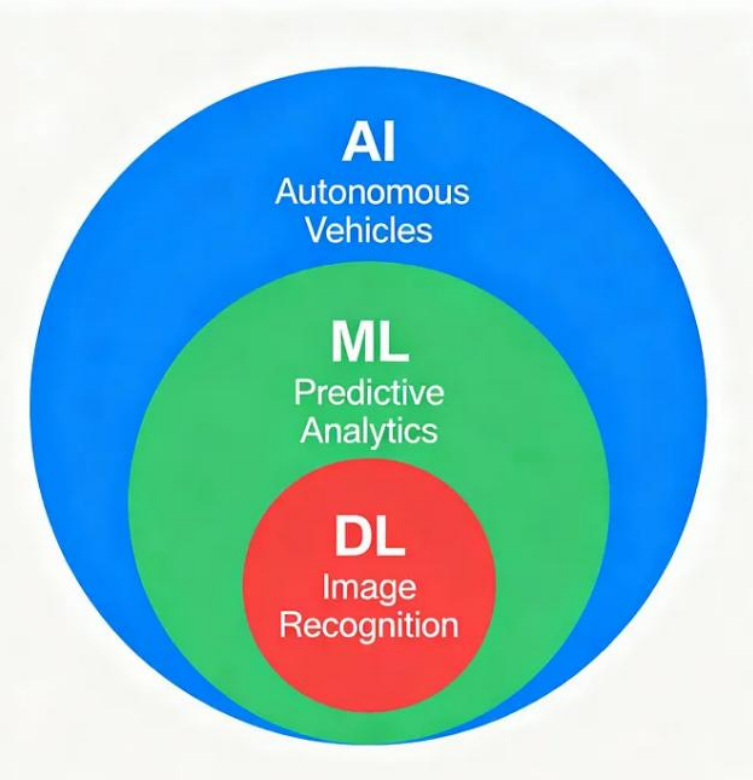
**Deep Learning**

Deep Learning is a subset of Machine Learning that specifically uses artificial neural networks (inspired by how the human brain works) with multiple layers to learn from data. Most specialized but most powerful for complex tasks. Requires massive amounts of data and computational power.

Instead of manually engineering features (like “pointy ears” or “whisker length”), deep learning automatically discovers the features needed for the task. DL is ML that uses brain-inspired neural networks. It’s the most “human-like” learning approach.

Real world example:

- Face recognition: Unlocking your phone with your face (uses deep learning)
- Language models: ChatGPT, Claude, and other large language models
- Image generation: DALL-E, Midjourney creating images from text
- Voice synthesis: Digital assistants sounding natural
- Autonomous vehicles: Deep learning helps process camera feeds and make driving decisions
- Medical imaging analysis: Detecting tumors and anomalies in CT scans




Visual Hierarchy between AI, ML and Deep Learning

* AI is the universe — everything intelligent
* ML is a planet in that universe — the intelligent things that learn
* DL is a city on that planet — the learners that use neural networks

### 3.0 Machine Learning Frameworks

Machine learning frameworks are tools, libraries, and resources that provide a standardized way for developers and other professionals to tailor the machine learning algorithm’s functionality without compromising the underlying code. The frameworks provide a simpler user interface that is accessible to many professionals,

instead of making it necessary to know and read code. Most of these are Python machine learning frameworks, primarily because Python is the most popular machine learning programming language.

Notable popular machine learning frameworks include:

1. **Scikit-learn**: A Python library that supports both unsupervised and supervised learning. If you're new to machine learning, scikit-learn is a great choice. It's effective for predictive data analysis and feature engineering.

2. **PyTorch**: A customizable option that uses building classes. If you're a Python developer searching for a framework with a shorter learning curve, PyTorch, when stacked against other frameworks, might be the one for you. Additionally, it's an open-source, deep-learning framework.

3. **TensorFlow**: A popular end-to-end machine learning platform that offers feature engineering and model serving. Finally, if you need a framework with robust scalability that works across a wide range of data sets, TensorFlow is a good choice.

Machine learning frameworks use many machine learning and deep learning functions, including time series, natural language processing, and computer vision. They can primarily help users deal with complex, challenging algorithms by making them more accessible and easier to use. machine learning frameworks expedite processes such as data visualization, model development, testing and logging, and data engineering.

**Benefits of Machine Learning Frameworks**

More specifically, one benefit of using a machine learning framework includes simplifying machine learning algorithms, which makes the technology much more accessible to businesses across different industries without needing to employ professionals trained in deep learning programming languages. This allows your business to discover insights and make data-driven decisions that would otherwise be inaccessible. To uncover these insights, machine learning frameworks can process vast amounts of data, also known as scalability, to identify patterns and reach conclusions.

Knowing where your machine learning algorithms are sourcing their information is also important. Not all data is usable or accurate, and you'll need to ensure that the sources are transparent, unbiased, and valuable. Otherwise, you risk extrapolating results that aren't actually true, which might lead your business astray with inaccurate conclusions. It's important to consider what exactly you're hoping to get out of machine learning and then take the time to research the different types of frameworks to ensure you're choosing the best one for your business.